

Functional Annotation Report: IscA (Q88PK6, PP_0844) in *Pseudomonas putida* KT2440

1. Summary / Answer to the Research Question

IscA (gene *iscA*, locus PP_0844; UniProt Q88PK6) is an **A-type carrier (ATC) protein of the ISC (Iron-Sulfur Cluster) biogenesis machinery**. Its primary function is **not** enzymatic catalysis but **molecular carriage**: it is a soluble cytoplasmic Fe-S cluster / iron chaperone that binds a pre-assembled iron-sulfur cluster (a labile, solvent-exposed [2Fe-2S], which it can convert/deliver as [4Fe-4S]) and **transfers it from the de novo assembly scaffold (IscU) to specific apo-target proteins**, thereby maturing them into active Fe-S enzymes. It belongs to the HesB/IscA family and carries the family-defining triad of conserved cysteines that ligate the cluster.

The gene identity is **confirmed and unambiguous**: the symbol *iscA*, the "Iron-binding protein IscA / Iron-sulfur cluster assembly protein" description, the HesB/IscA family, and the ATAP/FeS-cluster-insertion domains all coincide with a large, directly applicable body of literature on bacterial (chiefly *E. coli*) IscA. The 107-aa Q88PK6 sequence retains all three functional cysteines.

2. Molecular Function (Primary Role)

IscA is one of a small set of **A-type carrier proteins (ATCs)**. In the well-studied *E. coli* paradigm there are four ATCs — IscA, SufA, ErpA and the degenerate ATC* domain of NfuA — which "were shown to partially replace each other in their roles in [4Fe-4S] cluster insertion into specific target proteins" (Hasnat et al. 2021,).

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- **What it does:** receives an Fe-S cluster from the biosynthetic scaffold and hands it off to acceptor apoproteins. It does **not** synthesize sulfide or perform de novo cluster assembly;

that is the job of the upstream cysteine desulfurase IscS and scaffold IscU. The ISC system assembles a "transient formation of a 2Fe2S cluster on a scaffold protein from sulfide and iron salts" which is then transferred via carriers to acceptor apoproteins (Bonomi et al. 2022,

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- **Substrate / ligand specificity:** the "substrate" is the iron-sulfur cluster itself (and/or ferrous/ferric iron). UniProt annotates Q88PK6 as an **iron-binding protein**, and IscA family members are documented to bind iron as well as Fe-S clusters, supporting a dual role as an **iron chaperone/donor** that recruits iron for cluster assembly and as a **cluster carrier** for downstream delivery.

Iron-chaperone role and the "scaffold vs. carrier" resolution

Direct biochemical/genetic work (the Ding laboratory) clarified IscA's precise role. In vivo, "IscA binds iron" whereas the scaffold IscU and the frataxin homolog CyaY do not, and IscA "may act as an iron chaperone for the biogenesis of iron-sulfur clusters in *E. coli* cells under aerobic conditions" (Wang et al. 2010,

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Critically this activity is **oxygen-regulated**: "the strong iron-binding activity of IscA is greatly diminished in *E. coli* cells under anaerobic growth conditions," and IscA + its paralogue SufA are essential for Fe-S assembly aerobically but not anaerobically

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In vitro, with IscS/L-cysteine present, "iron-sulfur clusters are assembled in IscU, but not in IscA, suggesting that IscU is a preferred iron-sulfur cluster assembly scaffold protein," while in the absence of cysteine "nearly all iron is bound to IscA," which prevents precipitation of biologically inaccessible ferric hydroxide (Yang, Bitoun & Ding 2006,

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Interpretation for the target: IscA is best described as an **iron chaperone / cluster carrier acting downstream of the IscU scaffold**, keeping iron soluble and bioavailable under oxidizing conditions. Because *P. putida* KT2440 is an **obligate aerobe**, this aerobic iron-chaperone function is expected to be particularly relevant for PP_0844. - **Clients:** IscA/ATCs deliver clusters to numerous [4Fe-4S] enzymes, including respiratory-chain and metabolic proteins. Concrete examples of IscA-dependent maturation include subunits of the membrane-associated formate hydrogenlyase (FHL) complex — "The A-type carrier (ATC) proteins IscA

and ErpA probably deliver [Fe-S] to specific apoprotein components of the FHL complex because mutants lacking either protein exhibited strongly reduced hydrogen production. Neither ATC protein could compensate for the lack of the other" (Pinske et al. 2013,

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— and radical-SAM / molybdenum-cofactor and nitrate-reductase pathway proteins (Hasnat et al. 2021,

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Structural basis of function

The first Fe-S-cluster-bound structure of an IscA (from *Thermosynechococcus elongatus*) showed that "one partially exposed [2Fe-2S] cluster is coordinated by two conformationally distinct IscA protomers ... with asymmetric cysteinyl ligation by **Cys37**, **Cys101**, **Cys103**" and that the protein "carries three strictly conserved cysteine residues" (Morimoto et al. 2006,

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The **solvent-exposed** cluster and asymmetric, labile ligation are precisely what enable cluster handoff to acceptor proteins.

Bioinformatic confirmation for the target: Direct analysis of Q88PK6 (107 aa) identifies exactly three cysteines — **C35** within the conserved `...RTTG**C**SGL...` motif (the InterPro IPR017870 FeS-cluster-insertion conserved site) and **C99/C101** in the C-terminal `...RGE**C**G**C**GES...` motif. These are positionally homologous to the cluster-ligating Cys37/Cys101/Cys103 of *T. elongatus* IscA. Retention of this triad distinguishes a functional carrier from the "degenerate" A-type carrier domain (ATC) *lacking Fe-S cluster co-ordinating Cys ligands*" found in *NfuA* subfamily II (Py et al. 2012,

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— i.e., *P. putida* IscA is a bona fide, cluster-competent* ATC.

3. Biological Process / Pathway

IscA operates within the **ISC iron-sulfur cluster biogenesis pathway**, encoded by the *isc* operon (`iscR-iscS-iscU-iscA-hscB-hscA-fdx-iscX`).

Organism-specific genomic context (this study). A UniProt/KEGG query for *P. putida* KT2440 confirms the ISC cluster is present and canonically ordered around the target: **PP_0841 *iscR*** (2Fe-2S IscR transcriptional dual regulator) → **PP_0842 *iscS*** (cysteine desulfurase, EC 2.8.1.7) → **PP_0843 *iscU*** (scaffold) → **PP_0844 *iscA*** (target, Q88PK6) → **PP_0845 *hscB*** (co-chaperone) → **PP_0846 *hscA*** (Hsp70 chaperone) → **PP_0847 *fdx*** (2Fe-2S ferredoxin). A second cysteine desulfurase, *iscS-II* (PP_2435), lies elsewhere. This exact synteny with the *E. coli* `iscR-iscSUA-hscBA-fdx` operon is strong organism-specific evidence that PP_0844 functions as the A-type carrier/iron chaperone of a fully intact ISC machinery, co-regulated with the desulfurase, scaffold, chaperones, ferredoxin, and the IscR master regulator. Genetic dissection of this operon in *E. coli* placed IscA in the **assembler/delivery** stage: "the assembly of Fe-S clusters [is executed] by assembler machinery comprising of the `iscUA-hscBA-fdx` gene products," with IscS providing sulfur via cysteine desulfurase activity (Tokumoto & Takahashi 2001,).

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Coexpression of the whole `iscS-iscU-iscA-hscB-hscA-fdx` cluster dramatically increases holo-ferredoxin production, confirming these products "are involved in the assembly of the Fe-S clusters in a wide variety of Fe-S proteins" (Nakamura et al. 1999,).

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Pathway position (order of events): `IscS (Cys → sulfur) → IscU scaffold (+ Fe, + Fdx electrons, HscA/HscB chaperones) builds [2Fe-2S] → IscA/other ATCs receive & traffic the cluster (and supply iron) → apo-target proteins matured to holo-enzymes.`

Redundancy and regulation. The cell maintains an interconnected **carrier network** ("ErpA, IscA, SufA, and NfuA," plus the glutaredoxin GrxD) that bridges the housekeeping **ISC** and stress-responsive **SUF** systems (Fisher et al. 2024,).

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Among ATCs, "ErpA is the only ATC essential under aerobiosis," while IscA and SufA are partially redundant (Mandin et al. 2016,).

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This redundancy explains why *iscA* single mutants show only a "small but appreciable" defect in Fe-S enzyme activity (Tokumoto & Takahashi 2001,).

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even though IscA has non-redundant roles for specific clients (e.g., FHL; Pinske et al. 2013,).

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IscA is also reported dispensable under anaerobic conditions in an ISC-only background (Tanaka et al. 2016,

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consistent with condition-dependent redundancy.

Reduced redundancy in *P. putida* (this study). Database searches of KT2440 reveal a Fe-S biogenesis complement that is notably simpler than *E. coli*'s: the organism has the complete ISC operon plus the ATC carriers **ErpA (PP_0433)** and **NfuA (PP_2378)**, but **no canonical SUF operon and, critically, no SufA paralogue** (searches for *sufA/sufB/sufD/sufS/sufE* return no cognate SUF genes). Consequently IscA is the **sole IscA/SufA-type A-type carrier** in *P. putida*, so its aerobic iron-chaperone/cluster-carrier role is **less buffered** than in *E. coli* (where SufA can partially substitute). This suggests a comparatively more prominent, less redundant role for PP_0844 in this obligate aerobe — an inference from annotation searches that merits direct experimental verification.

The end products of the pathway — mature Fe-S proteins — are essential for respiration, the TCA cycle (aconitase), isoprenoid biosynthesis (IspG/IspH), DNA repair, and gene regulation, so IscA contributes indirectly but importantly to central metabolism.

4. Subcellular Localization

IscA functions in the **bacterial cytoplasm**. It is a soluble protein with no signal peptide or transmembrane segment (107 aa, consistent with the compact soluble ATC fold), and it acts on cytoplasmic scaffolds (IscU) and cytoplasmic/inner-membrane-associated apoprotein clients. Its membership in the cytoplasmic Fe-S trafficking network (Fisher et al. 2024,

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situates its activity in the cytosol, where it can also deliver clusters to membrane-embedded complexes (e.g., FHL) from the cytoplasmic face. In *P. putida* this network comprises the ISC system plus the carriers IscA, ErpA and NfuA (no SUF pathway; see §3).

5. Evidence Summary

Claim	Evidence type	Source
IscA is an A-type carrier delivering [4Fe-4S] to targets	Experimental (genetics, <i>E. coli</i>)	P 33782054
Three conserved Cys ligate an exposed [2Fe-2S] across an asymmetric dimer	Experimental (X-ray structure, <i>T. elongatus</i>)	P 16730357
Q88PK6 retains the three functional cysteines (C35, C99, C101)	Bioinformatic (sequence analysis, this study)	UniProt Q88PK6 + P 16730357
IscA acts downstream of IscS/IscU in the <i>isc</i> operon	Experimental (operon mutagenesis)	P 11432781 ; 10393315
Non-redundant delivery to specific clients (FHL)	Experimental (deletion phenotypes)	P 23558265
Partial redundancy with SufA/ErpA; ErpA essential aerobically	Experimental / review	P 27651365 ; 26560204
Part of cytoplasmic ISC↔SUF carrier network incl. GrxD	Experimental (chemoproteomics)	P 38944118
Distinct from degenerate ATC* (lacking Cys) of NfuA	Experimental / phylogenomic	P 22966982
IscA is an iron chaperone; iron binding is oxygen-dependent (aerobic)	Experimental (in vivo + in vitro, <i>E. coli</i>)	P 20942799 ; 16877383
IscU (not IscA) is the preferred de novo cluster scaffold	Experimental (in vitro reconstitution)	P 16877383
PP_0844 sits in a complete, canonically ordered <i>P. putida</i> <i>isc</i> operon (PP_0841–0847)	Bioinformatic (genomic synteny, this study)	UniProt/KEGG; P 11432781
<i>P. putida</i> has ISC + ErpA/NfuA but no SUF operon or SufA paralogue → IscA less redundant	Bioinformatic (annotation search, this study)	UniProt/KEGG; P 38944118

6. Supported and Refuted Hypotheses

Supported - H1: IscA is an Fe–S cluster carrier/chaperone in the ISC pathway (not a catalytic enzyme). ✓ - H2: *P. putida* IscA retains the functional cluster-ligating cysteine triad. ✓ (sequence analysis) - H3: IscA acts in the cytoplasm and delivers clusters to specific apoproteins with partial redundancy among ATCs. ✓

Refuted / not supported - That IscA performs de novo cluster synthesis or sulfur mobilization — those are IscS/IscU functions; IscA is downstream. ✗ - That IscA is individually essential — single *iscA* deletion has only a modest phenotype due to ATC redundancy. ✗

7. Limitations and Future Directions

- Nearly all direct functional/structural data derive from *E. coli* and cyanobacterial orthologs; there is little *P. putida*-specific experimental characterization of PP_0844. Functional assignment for Q88PK6 is by strong homology and conserved-motif inference rather than direct in-organism assay.
 - The long-debated "scaffold vs. iron-chaperone vs. cluster-carrier" emphasis of IscA is condition- and organism-dependent; the precise balance in *P. putida* (an obligate aerobe with active oxidative metabolism) is untested.
 - **Future work:** (i) *P. putida* Δ *iscA* / Δ *sufA* / Δ *erpA* phenotyping under aerobic vs. iron-limited stress; (ii) in vitro reconstitution to confirm Fe/[2Fe–2S] binding by recombinant Q88PK6; (iii) identify *P. putida*-specific client apoproteins (e.g., ferredoxins, aconitase, IspG/IspH) that depend on IscA.
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Gene identity verification (as required)

- Symbol **iscA** ✓ matches "Iron-binding protein IscA / Iron-sulfur cluster assembly protein."
- Organism ✓ *Pseudomonas putida* KT2440 (OX=160488), locus PP_0844.
- Family/domains ✓ HesB/IscA family; ATAP core (IPR000361), FeS-cluster-insertion conserved site (IPR017870), HesB-like sf (IPR035903) — all consistent with the literature above.

- No conflicting "same-symbol, different-gene" ambiguity was encountered; the eukaryotic "ISCA/MagR" magnetoreception literature (e.g.,

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) refers to homologous ISCA proteins and does not contradict the bacterial IscA carrier function assigned here.

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